F FWallSim · Firewall Course

CLASS 12

# SD-WAN with two ISPs

SD-WAN members, performance SLA, failover

Advanced | ~30 minutes | FWallSim v1.0.0 (FortiOS 7.4 style)

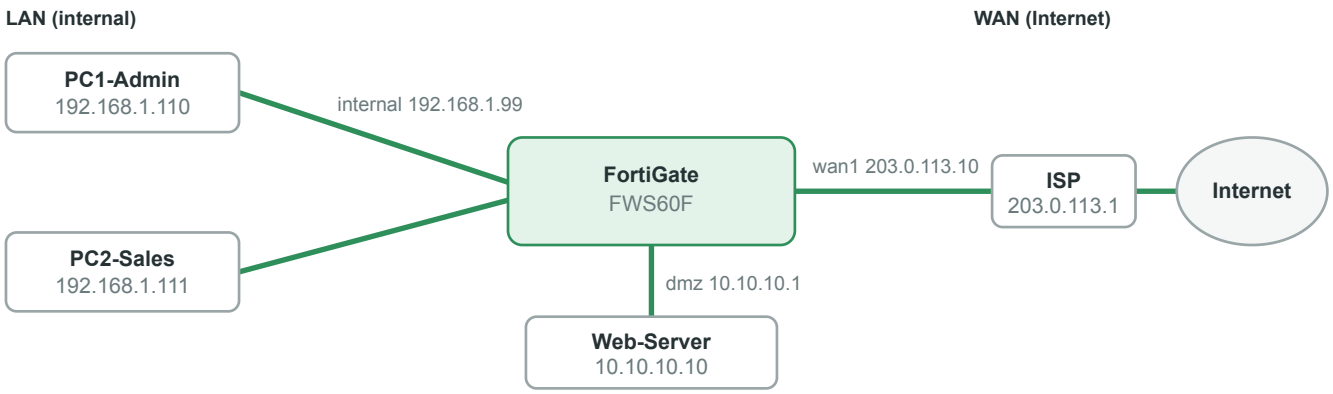
### YOU WILL LEARN

- SD-WAN members
- performance SLA
- failover

### BEFORE YOU START

Finish Class 11 first.

## The lab network




## The goal

### THE IDEA

A second internet line (ISP2) is now plugged into wan2. Build SD-WAN so the office survives an ISP1 outage automatically.

## How to start

1. Open FWallSim and log in.
2. Click **Labs** in the header.
3. Find **SD-WAN with two ISPs** and click **Start**.
4. The Lab Guide panel opens. Each task below gets a green tick as soon as it works.

## Tasks

1. **SD-WAN is enabled with wan1 and wan2 as members**

   1) Network > SD-WAN: enable SD-WAN and add wan2 (gateway 198.18.10.1). 2) Firewall Policy: change policies 1 and 3 (Outgoing) and policy 2 (Incoming, the website) from wan1 to virtual-wan-link. FortiOS will not add an interface to SD-WAN while policies use it directly. 3) Back in SD-WAN, add wan1 (gateway 203.0.113.1).

2. **Policy 1 goes out the SD-WAN zone and a default route points to it**

   Edit policy 1: Outgoing interface virtual-wan-link. Static Routes: 0.0.0.0/0, Interface = SD-WAN zone virtual-wan-link.

3. **Internet-User can still reach the website**

   Quick test "Internet user visits your website" (before switching on the ISP1 outage).

4. **A Performance SLA probes an internet server (not the ISP gateway)**

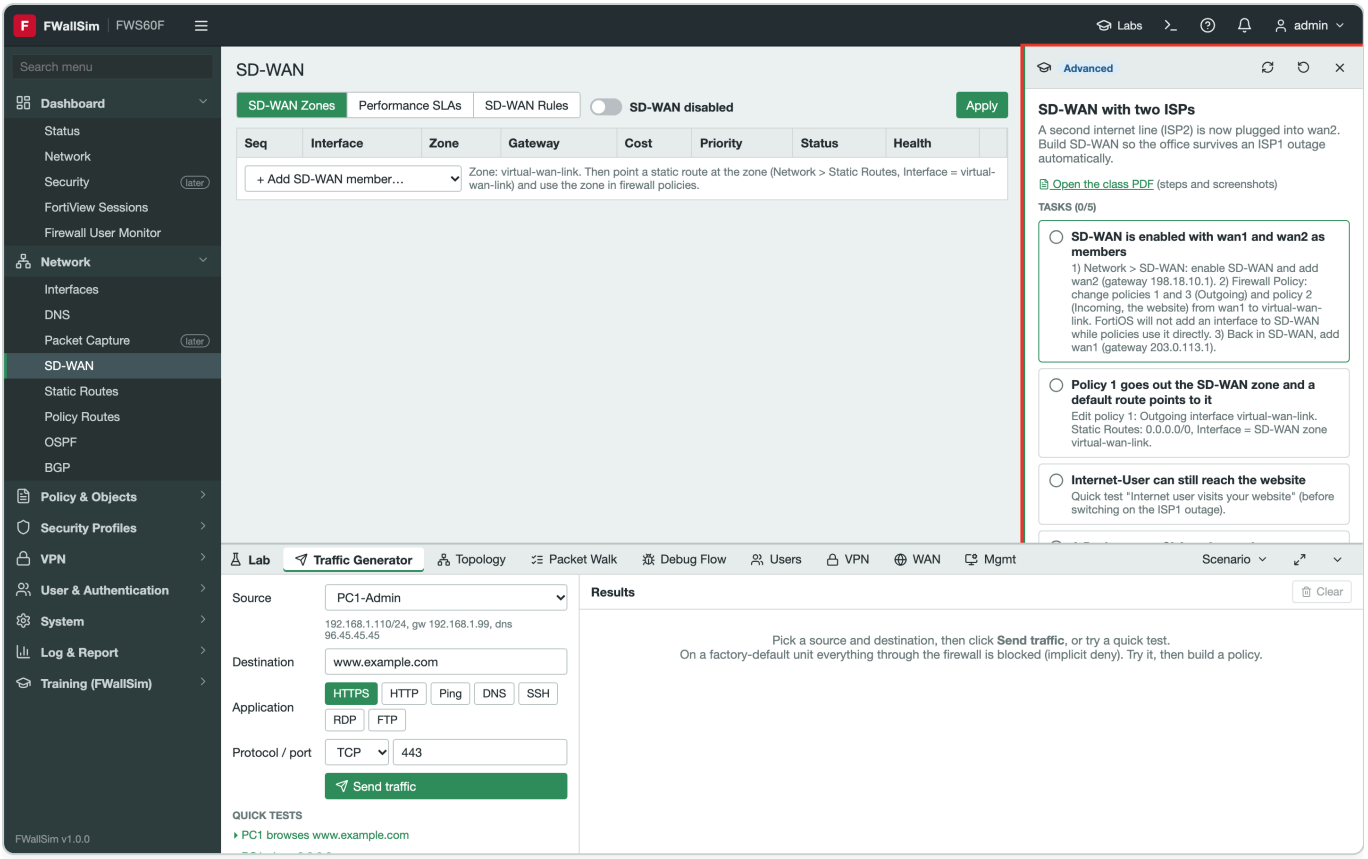
   Performance SLAs > Create New: server 8.8.8.8, both members.

5. **With an ISP1 outage, PC1 still reaches 8.8.8.8**

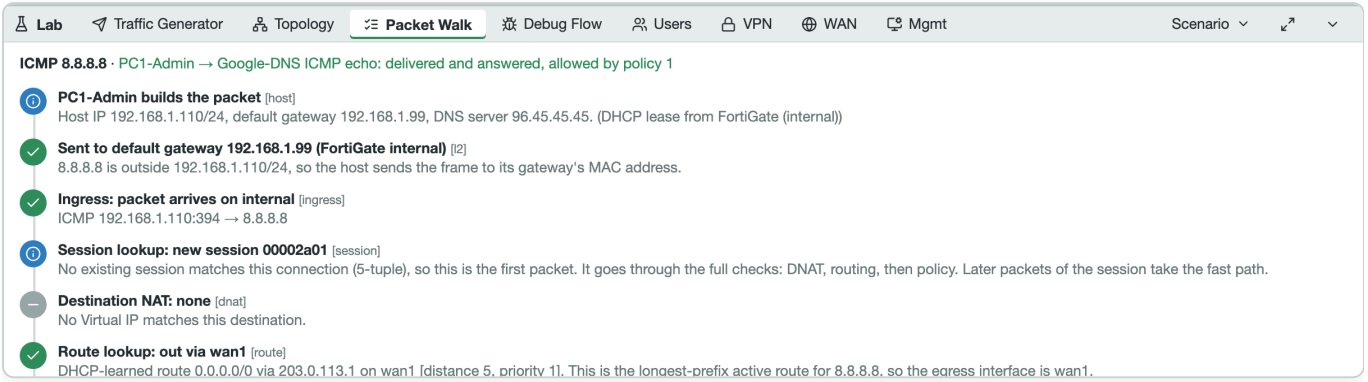
   Lab > WAN: switch on "ISP1 outage", then ping 8.8.8.8 from PC1.

## Screenshots

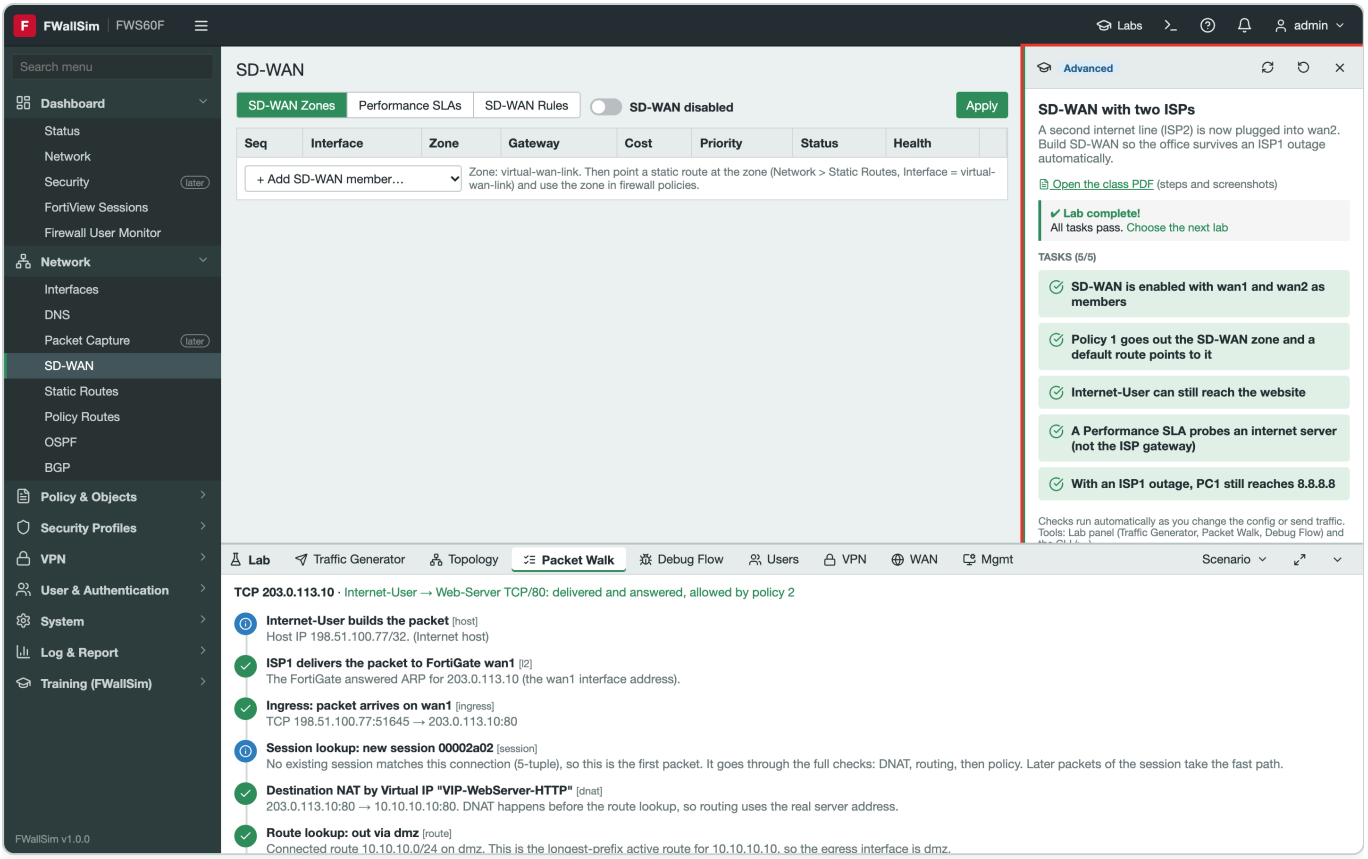
Numbered red markers show where to click or what to look at.



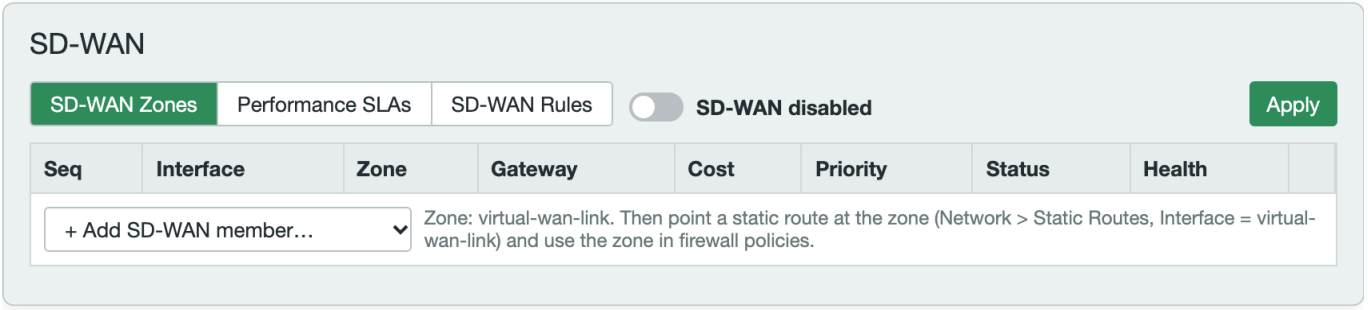

*Lab Started*



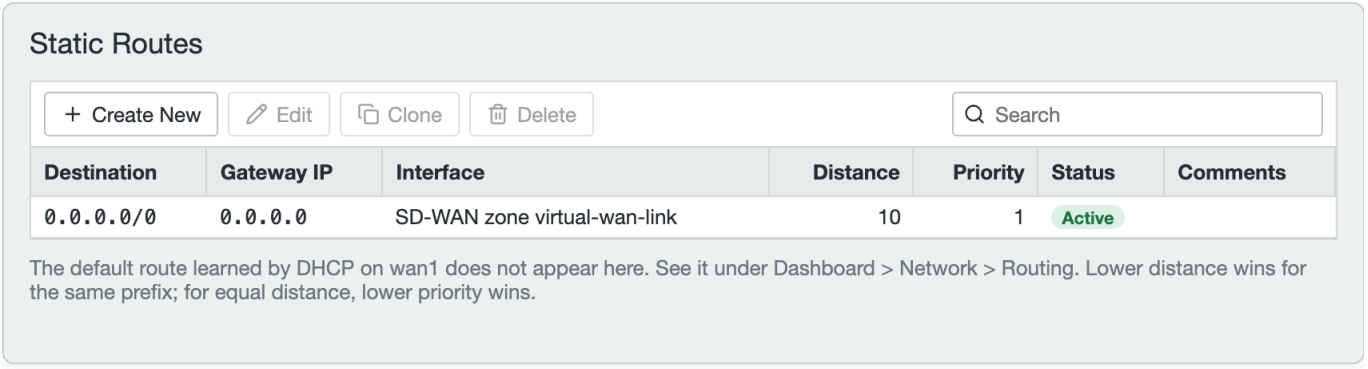

*Packet Walk Before*




*Lab Solved*




*SD-WAN*




*Static Routes*




*Packet Walk After*

**Next:** Class 13: Dynamic routing: OSPF over VPN and BGP